

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2022

Bachelor in Computer Applications

Course Title: **Mathematics I**

Code No: Semester: I

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. In a certain village in Nepal, all the people speak Nepali or Tharu or both languages. If 90% speak Nepali and 20% speak Tharu, how many people speak: i) Nepali language only ii) Tharu language only iii) both languages
3. If $(x - ty = \frac{5 - 6i}{5 + 6i})$, prove that $(x^2 + y^2 = 1)$.
4. Define a function. Show that the function $(f: \mathbb{R} \rightarrow \mathbb{R})$ defined by $(f(x) = 3x + 5)$ is bijective.
5. If (A) is the A.M. and (H) is the H.M. between two numbers (a) and (b) , show that: $(\frac{a - A}{b - A} = \frac{A - H}{H - A})$
6. Define matrix. If $(A = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix})$ and $(B = \begin{pmatrix} -2 & 1 \\ 3 & 2 \end{pmatrix})$, show that: $(AB)^T = B^T A^T)$.
7. Prove that: $(\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (a - b)(b - c)(c - a))$
8. Find the eccentricity and foci of the ellipse: $(25x^2 + 4y^2 = 100)$.

Group C

Attempt any TWO questions [2x10=20]

9. a) A bag contains 8 red balls and 5 blue balls. In how many ways can 3 red balls and 4 blue balls be drawn? b) Find the volume of the parallelepiped whose concurrent edges are represented by the vectors $(\vec{i} - 2\vec{j} + 3\vec{k})$, $(-3\vec{i} + 4\vec{j} - 5\vec{k})$, and $(\vec{i} + 2\vec{j} - 3\vec{k})$.
10. a) Find the Taylor Series expansion of $(f(x) = x^3 - 2x + 4)$ at $(a = 2)$. b) In how many ways can the letters of the word 'CALCULUS' be arranged so that the two C's do not come together?
11. Define exponential and logarithmic functions. If $(f(x) = \log \frac{1 - x}{1 + x})$ for $(-1 < x < 1)$, show that: $(f(\frac{2ab}{1 + a^2 b^2}) = 2f(ab) \text{ where } |ab| < 1)$